

# Is the tokenization of property the next step in the financialization of housing?

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## ABSTRACT

While financial incumbents hail the transformative potential of tokenization, critical research in economic geography and housing studies argue that the digitalization of real estate and the rise of blockchain-based platforms are more likely to usher in a new round of reintermediation, deepening financialization. Yet empirical engagement with real-world examples remains limited, leaving open the question of how housing tokenization operates in practice. This article addresses that gap by examining a European startup that capitalized on early regulatory openings to tokenize residential properties. The project sought to extend traditional securitization practices by outsourcing blockchain infrastructure, aiming to expand access to property investment and offer developers new funding channels. Drawing on qualitative research methods, we explore the mechanics, aspirations, and the limits of housing tokenization. The case exposes the tension between the rhetoric of blockchain's extension and democratization of securitization and the persistent structures of conventional finance. Far from transforming real estate markets or forging a new frontier of financialization, tokenization here reconfigures familiar arrangements in a technical guise, albeit with modest participatory and community-facing elements. As regulatory frameworks mature and financial incumbents consolidate their positions, careful empirical scrutiny is required to assess what tokenization does, and does not, change on the ground.

## 1. Introduction

Blockchain technology is transforming one of finance's most conservative domains: real estate investments. At the center of this transformation is the practice of converting real-world assets (RWAs) into tokens - digital representations of ownership or economic rights in assets such as real estate, debt, or equities (Stabile et al., 2020; Wandmacher & Wegmann, 2020; BIS, 2024). This process, known as tokenization, is expected to enable individuals to buy and sell indirect ownership in high-value, income-generating properties for as little as \$10, with the added benefit of near-instant secondary market trading (Chow & Tan, 2022). As Larry Fink put it in his 2025 Annual Chairman's Letter to investors, tokenization is the way "to finish the market democratization that began 400 years ago and let more people own a meaningful stake in the growth happening around them."<sup>3</sup> But is it really?

This article takes a critical look at the conceptual and empirical

questions raised by blockchain's adoption in real estate, notably the tokenization of housing. It starts from the assumption that understanding where tokenization is headed requires attention to the conjunctural settings shaping its integration into the conventional real estate world, as well as the perspectives and experiences of those actively "practicing" it. This matters because housing isn't just another asset, it's both structurally significant and politically sensitive in many advanced economies. After all, homeownership often represents the largest investment for families and serves as a form of alternative welfare globally. However, the growing trend of treating homes and land primarily as objects of speculative lending and investment, separating them from productive growth and incomes, has made the interactions between land, property, and the financial system key factors shaping the macroeconomy (Ryan-Collins et al., 2017, p.110).

Economic and urban geographers describe these dynamics using the concept of financialization, a process of capital accumulation in which

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“profit-making occurs increasingly through financial channels rather than trade and commodity production” (Krippner, 2005). In the housing sector, financialization operates on both the supply and demand sides of housing provision. On the supply side, it manifests through the proliferation of new financial actors and real estate funding mechanisms that channel global financial capital into local housing markets (Aalbers, 2017), while also facilitating the assetization of housing rights so that they can be capitalized as future revenue streams configured as financial instruments (Birch & Ward, 2024). On the demand side, it manifests in the financialization of property ownership, as owner-occupiers, investors and state actors increasingly treat homes as revenue-generating assets for investment, speculation, and wealth accumulation rather than as a fundamental right (Aalbers, 2017; Çelik, 2024; Fields, 2018; Leijten & de Bel, 2020; Mazzucato & Farha, 2024).

Arguably, real estate tokenization can affect both sides of housing provision and further drive the financialization of housing, because it not only shares many features with traditional securitization but also can work as its “extension” (Wandmacher & Wegmann, 2020, p.178). For decades, securitization has served a dual purpose: connecting capital users (borrowers) with a broader and more diverse pool of capital providers, including pension funds, insurance companies, and other institutional investors, and enabling the extraction of value from illiquid assets, improving liquidity, and facilitating capital reallocation, thereby reducing the constraints associated with direct investments (Baker et al., 2019; Kreppmeier et al., 2023; Stabile et al., 2020). More importantly, it allowed issuers to transfer risks to investors. Now, tokenization, being more flexible and offering greater control, is expected to upgrade securitization by increasing efficiency and removing intermediaries from the process (Wandmacher & Wegmann, 2020).

It is important to keep in mind, however that securitization is the practice that underpins the emergence of residential Real Estate Investment Trusts (REITs), financial vehicles that played a key role in transforming housing into a mainstream investment asset class and deepened housing financialization (Wijburg et al., 2018). As Taylor and Aalbers (2024) show, when pooling diverse forms of housing, that may well include mobile homes, student accommodation, or even socially rented apartments, managers of such vehicles tend to prioritize profit over affordability, thereby exacerbating housing inequalities and expanding what the authors call “territories of dispossession.” Blockchain-powered platforms can be designed to do the same: functioning as what Crandall (2023) describes as “digital architectures of exclusion”, rather than tools for democratization, they can be instrumental in extending exclusionary dynamics already present in earlier forms of real estate financialization (Liu, 2025). This raises further concerns regarding the tokenization of housing.

Convinced that the political implications of housing tokenization demand critical attention, we seek to address the following question: how does the tokenization of housing operate in practice and to what extent could blockchain reshape the way finance penetrates residential real estate? In other words, our primary focus is not on housing financialization per se, but on critically examining the mechanisms of housing tokenization as they are realized in practice.

To do so, we focus on the case of Luxembourg, a leading center for securitization and structured vehicles in the Euro area, ranked third after Ireland and Italy in terms of securitized asset volume (PWC, 2023). There, a private startup leveraged newly implemented national regulations to raise funds for the acquisition of a multi-dwelling property under construction by using a blockchain-based platform to conduct an initial securities offering to retail investors. The platform enabled the issuer to lower the entry threshold for private equity investment, but did not attract investors with a profile that differed significantly from more ‘traditional’ buy-to-let investors. While token holders received rights to a share of future rental income, voting rights on future developments, and the option to become tenants, regulatory changes limited the liquidity of the investments. Given the initial reliance on a bank “bridge” loan to jump start the venture as well as the platform’s perpetuation, in

practice, of the clear distinction between investor, owner and tenant, we thus question the extent to which tokenization can develop into a new mode of housing provision.

As real-life examples of housing tokenization remain limited, we find value in documenting the attempt by real estate professionals in Luxembourg to leverage blockchain as a means of navigating challenges posed by shifting market conditions. While the venture ultimately faced mounting obstacles, its development offers more general insights into the direction and scope of housing tokenization, and provides reasons to remain cautious about its ‘democratizing potential’, at least in this phase of the tokenization-enabled digital transformation of capital markets. With this in mind, this article is organized in three main parts. We start by reviewing the emerging literature on the tokenization of real estate, taking two complementary perspectives. From the perspective of economic geography, we first situate tokenization in relation to securitization and the development of blockchain. Then, from the perspective of urban geography and housing studies, we show how the tokenization of housing could in practice constitute a new mode of housing provision. We then discuss the methodology used to research a particular deployment of blockchain in the real estate sector, a venture deeply affected by a rapidly changing regulatory environment. The third main section begins by introducing this venture, then critically examines its implementation on the ground in relation to the claims made about the potential of tokenization.

## 2. Tokenization: an unruly offspring of securitization and blockchain

Before empirically engaging with real estate tokenization as it unfolds in Europe, we first examine the existing literature on tokenization to better understand its mechanisms and its promise to broaden retail investor’s access to finance by extending securitization. We then situate these developments within the broader history of housing provision, considering how tokenization may complement or extend existing securitization practices that have increasingly been used to address constraints in retail banking.

### 2.1. Tokenization of real estate: unlocking more of property investment potential?

Until recently, tokenization was often used as a broad umbrella term that lacked specificity (Stabile et al., 2020). Simply put, when people referred to “tokens,” it wasn’t always clear what kinds of assets or processes they were talking about. Part of this ambiguity stemmed from the fact that tokenization initially developed alongside decentralized finance or DeFi, a movement that positioned itself as an alternative to traditional finance, or even a way to bypass its core intermediaries, like banks (Bok, 2024). Consequently, blockchain, the technological innovation behind tokenization, was often framed as a form of *institutional innovation* (Davidson et al., 2018) that would enable a bonfire of new assets whose relation to existing financial instruments remained poorly understood (Sazandrishvili, 2020).

Now, following a series of rulings and statements by the US Securities and Exchange Commission on tokenization, which increasingly link tokenized ventures to securitization, together with new regulations in the EU, such as Markets in Crypto-Assets Regulation (MiCAR), we can observe a clear shift toward convergence (Proskurovska & Birch, 2025). In these emerging regulatory framings, blockchain often recedes into the background (cf. BIS, 2024). Yet, an understanding of the interplay between securitization and blockchain-enabled tokenization is essential in order to make sense of tokenization, particularly in relation to the trading of fractionalized real estate.

Securitization can be defined as a method used to generate cash flow from financial assets and other rights such as the ownership of real estate, art, etc. (Schwarcz, 2022). In practice it involves financial transactions in which income-producing assets, such as rental property, or

loans, are pooled and their future cash flows are sold to a trust or other special purpose vehicle (SPV) (Schwarcz, 2012). The SPV then issues securities backed by these assets for investors to buy. Simply put, securitization is a product of financial globalization: it is a key mechanism for the redistribution of credit and liquidity risks from issuers to investors best equipped to manage it while allowing them to earn income through rental yields or mortgage interest without the complexities of direct property ownership (Buchanan, 2017). That is why professionals working in banking or real estate sectors often see securitization as a way to improve liquidity, facilitate capital reallocation, and reduce the constraints of direct investment. It also lowers capital costs for borrowers, improves tax efficiency, and provides investors with access to traditionally hard-to-reach asset classes (Baker et al., 2019; Kreppmeier et al., 2023; Stabile et al., 2020).

As noted earlier, REITs exemplify this approach. Introduced in the United States in 1960, they were designed to democratize access to real estate investment and avoid corporate-level taxation by mandating high dividend payouts. Today, REITs are widely regarded as liquid and transparent financial vehicles that allow both individuals and institutions to earn returns from real estate without the responsibilities of direct ownership. Whether publicly listed or private, these vehicles offer investors a way to earn real estate returns without the hassle of managing properties, enhancing liquidity despite the inherently fixed and local nature of real estate (Aalbers et al., 2023).

The existing real estate and financial literature often focuses on how securitization vehicles have enabled investors to bypass costly direct transactions, diversify portfolios, and manage risk (e.g., Baum, 2022). However, these benefits often come with trade-offs, with critics highlighting issues such as reduced transparency and increased systemic risk (Feng et al., 2011). For example, publicly listed REITs expose investors to governance mechanisms embedded within legacy financial infrastructures, such as stock exchanges, which exert structural power by regulating the essential systems underpinning financial markets (Petry, 2021). This setup also limits investor control over asset selection and portfolio strategy (Gupta et al., 2020; Mohamed, 2022; Mottaghi et al., 2024; Smith et al., 2019). As a result, those seeking greater control often turn to private real estate funds, which offer more discretion over investment decisions but usually come with high minimum capital requirements and limited access. Indeed, securitized real estate products, especially in private markets, are burdened by complex legal and regulatory compliance, due diligence, administrative overhead, and intermediary fees. For example, fulfilling stringent Know Your Customer (KYC) and Anti-Money Laundering (AML) requirements for identity verification and reporting are well known factors that increase onboarding complexity and create barriers for smaller investors (Schwarcz & Bourret, 2023).

In both cases, reliance on conventional financial infrastructures and intermediaries such as banks and brokers hinders REITs' competitive advantages by limiting possible economies of scale (Highfield et al., 2021). In addition, these costs do not only affect investors in equity; it can also limit access to financing for real estate developers. Securitization vehicles such as mortgage-backed securities or private debt funds could indeed offer a lower-cost alternative to traditional bank loans by pooling capital and distributing risk. But due to the high cost and complexity of structuring these instruments, only large-scale developers can typically access them. Small and mid-sized developers are often excluded, lacking the scale or capital required to participate, and are thus pushed toward more expensive or restrictive financing options (Varouchakis, 2024).

Tokenization offers an alternative approach to converting traditionally illiquid and costly assets into highly liquid and cost-effective ones (Fisch et al., 2022; Mohamed, 2022; Mottaghi et al., 2024; Proskurovska & Birch, 2025). It also can provide a viable funding source for poorly capitalized financial markets (Chow & Tan, 2022). Blockchain is the key technological innovation that makes the process of investing in and owning assets categorically distinct from traditional methods of

establishing relationships with real estate assets. It has been proven to create tamper-proof transaction records enabling real-time updates alongside a unified consensus mechanism, and thus to streamline the settlement and transfer of digitally represented value across industries, real estate included (Avci & Erzurumlu, 2023; Gupta et al., 2020; Kreppmeier & Laschinger, 2023; Lambert et al., 2022; Proskurovska & Morawski, 2026; Schwarcz & Bourret, 2023).

More importantly, blockchain offers a mechanism for significantly reducing transaction costs through the use of smart contracts (SCs). SCs are self-executing digital agreements with the terms directly encoded, allowing operations to occur automatically once predefined conditions are met (Tian et al., 2020). This automation enables the direct transfer of digital representation of RWA between parties, that is without the need for traditional intermediaries such as brokers, custodians, or notaries (Khan et al., 2021; Proskurovska & Morawski, 2026). Because these contracts are recorded on a tamper-proof blockchain ledger, the integrity of the agreement is maintained: no party can alter the terms unilaterally once the contract is deployed. That is why SCs can incorporate identity verification using credentials from trusted sources, making them compatible even with legacy financial infrastructure (Aránguiz et al., 2021; Proskurovska & Dörny, 2018). This enables compliance checks like KYC and AML to be built directly into the code, so only verified users can buy, hold, or transfer the asset (Garcia-Teruel & Simón-Moreno, 2021; Tokeny et al., 2023).

Representing a traditional security as a digital token in this way not only harnesses legacy infrastructures to enforce regulatory compliance automatically but also speeds up settlement, making transactions nearly instantaneous (BIS, 2024). Because blockchain networks update continuously, token holders can also access real-time information about asset value, usage, or even maintenance costs (Avci & Erzurumlu, 2023; Bellavitis et al., 2023). These embedded rules can also automate workflows like investor onboarding, compliance, and settlements, reducing reliance on intermediaries and improving overall efficiency (Avci & Erzurumlu, 2023; Baum, 2020; Saari et al., 2022). In such a blockchain-based system, token holders, no matter how large or small, could even vote on future investment strategies, offering a level of governance and participation rarely seen in conventional securitization. While this section has laid out the ways in which tokenization could potentially offer new opportunities, it is worth taking a more critical look at its potential impacts.

## 2.2. Tokenization: same game, new tools?

Economic geographers pay close attention to the role of digital and institutional innovations in reshaping economic coordination and urban landscapes. As digital technologies in general, and blockchain in particular, were hailed as disruptive tools meant to eliminate intermediaries and dismantle inefficient monopolies, a growing body of research remains cautious, arguing that disruptive technologies often become domesticated by incumbents, who use them to channel financial capital into parts of the built environment that were previously shielded from such flows (Fields, 2019; Langley & Leyshon, 2017; Proskurovska, 2023a, 2023b; Rella, 2019; Zook & Grote, 2016; Zook & Grote, 2020).

Take one notable example: profit-sharing crowdfunding. This mechanism was also introduced as an alternative to traditional funding models like debt, equity, donations, or rewards. Unlike traditional lenders such as banks and venture capitalists, it appealed to a broader, unspecified group of individuals (Bieri, 2015; Langley & Leyshon, 2017). This funding method was expected to eliminate intermediaries, reduce transaction costs, and foster a more democratic, bottom-up investment landscape (Bieri, 2015; Fitzmaurice et al., 2020). Yet, over time, it became evident that while crowdfunding platforms displaced traditional lenders, they did not eliminate intermediation but reconfigured it (Ertürk, 2016; Ertürk & Solari, 2007; Rella, 2019). Platform providers assumed intermediary roles, charging fees for their services and transforming crowdfunding into a revenue-generating

infrastructure (Langley & Leyshon, 2017). More importantly, as demonstrated by Lolli's (2019) analysis of commercial crowdfunding platforms, the process of reintermediation and capitalization, largely enabled by new digital infrastructures and central to the financialization of crowdfunding, has become less about fostering direct collaboration between consumers and producers and more about minimizing financial risks for large stakeholders traditionally outside crowdfunding platforms. Over time, platforms began accepting investments from hedge funds, money managers, banks, and insurance companies, who saw online loans as an alternative asset class to bonds for portfolio diversification. In some cases, platforms fund the loans themselves (balance sheet lending), turning crowdfunding portals into channels for traditional lending (King, 2023). Crowdfunding has thus evolved from peer-to-peer lending to *marketplace lending* (King, 2023) or a mechanism of risk transfer, where corporate players secure capital while backers shoulder uncertainty (Lolli, 2019).

Still, even though the blockchain-enabled tokenization of securities may appear to be just another step in the ongoing restructuring of finance, it nevertheless operates through somewhat distinct mechanisms of economic coordination (Davidson et al., 2018; Mora-Cruz & Palos-Sanchez, 2023; Proskurovska & Birch, 2025; Zook & Grote, 2020). Firstly, the circulation of blockchain-based assets relies on different economic communities, whose members are capable not only to govern formalization of tokenized securities, but also navigate infrastructures that enable the circulation of both conventional assets and tokenized securities (Breidbach & Tana, 2021; Proskurovska, 2023b, 2023a; Rella, 2020; M. Zook & Grote, 2016; M.A. Zook & Blankenship, 2018). Secondly, blockchain does more than simply reshuffle or eliminate intermediaries; it also can transform the processes through which digital and material spaces are integrated, by shifting the scale at which economies and finance operate (Zook & Grote, 2022). In other words, as blockchain enabled platforms reintermediate, they can both foster direct connections between consumers and producers of housing finance, and restructure global-local linkages, ultimately shifting financial risk from dominant incumbents to the rest of market participants (Proskurovska, 2023a, 2023b).

Before empirically engaging with a case of tokenization of housing, to examine whether housing tokenization follows a similar trajectory, we first need to shed more light on the very processes it can possibly affect, namely those happening at the local scale. We thus turn to the different modes that the provision of housing has taken in Western economies

### 2.3. Tokenization of housing and local housing dynamics: successor to buy-to-let or a new financial frontier?

Housing provision tends to operate as a black box, with the processes funneling large amounts of global capital into local housing markets unknown to many involved in local housing markets. To understand how the tokenization of real estate could impact housing provision, we situate this recent development within a general model of housing provision. Indeed, in western countries, we have identified a succession of different *modes of housing provision*, that is, configurations in which property developers and consumers (such as owner-occupiers or property investors) temporarily stabilize their interactions around a particular type of housing 'product' made possible by a particular regulatory and financial context. The notion of mode of housing provision emphasizes the interconnectedness of the 'demand' and 'supply' sides, subsuming them in a broader mode of governance. A shift from one mode of housing provision to another can be prompted by a growing incompatibility of either property developers or housing consumers with the housing product, or by changes in the policy or financial context that affect the opportunity structure.

The first of these modes, owner-occupation, can be understood as a package of housing services linked to investment vehicles through mortgages (Smith, 2009: 233), and heavily incentivized by many

governments. It marks the *horizontal expansion* of homeownership, which had until the middle of the twentieth century been reserved to the highest social strata. The spread of the owner-occupation mode of housing provision has however tied the stability and accessibility of housing to a broader apparatus of monetary, financial, regulatory and calculative techniques designed to sustain the futurity and desirability of [residential] property not as a specific plot of land or a building, but an abstract asset class (Ashton, 2018, p. 233). As a result, property and housing in particular, is deeply interconnected with financial markets (Allen & Yago, 2010; Ashton & Christophers, 2018; Sassen, 2009), and the act of acquiring a home in many advanced economies implies forging an intimate relationship between global financial markets, everyday life, and human labor which is inherently local (García-Lamarca & Kaika, 2016). The need to expand this market while sustaining the financial interests of property developers and property owners drives prices up, increasingly limiting homeownership to those with the greatest purchasing power (Arundel & Ronald, 2021).

The increasing stratification in relation to homeownership created a set of housing consumers who successfully navigated owner-occupation and started to accumulate multiple properties, stimulated by the deregulation of private rental markets and the introduction of fiscal subsidies for property investors (Mezaroş & Paccoud, 2022), as well as the creation of new financial assets such as the buy-to-let mortgage (Leyshon & French, 2009). Property developers in turn marketed part of their supply directly towards buy-to-let investors, who are able to access capital more easily and thus buy earlier, more quickly and in greater quantity than owner-occupiers (Mezaroş et al., 2025). This new mode of housing provision linking small-scale property investors to developers, and which marks an *upwards hierarchical move* in relation to owner-occupation, can be called the 'buy-to-let' mode. 'Buy-to-let' investments can occur through different channels, with important socio-spatial implications (Paccoud, 2017). They are linked to the development of build-to-rent models (Nethercote, 2020), developers marketing their supply to buy-to-let investors (Pollard, 2023), housing as a site of investment for the super-rich (Fernandez et al., 2016) or overseas investors (Ho & Atkinson, 2018) and the facilitation of short-term rentals such as through AirBnB (Cocola-Gant & Gago, 2021).

In response to rising property prices and constrained retail investment, some construction and development companies have begun applying a range of strategies typically used by financial institutions (Aalbers, 2018; Romainville, 2017). This included leveraging securitization mechanisms - such as selling equity in private REITs and SPVs - to raise capital, which can look like the repurposing of tools traditionally used by banks to fund mortgage lending (cf. Proskurovska, 2023a, 2023b), engaging with private equity investors buying into large rental housing portfolios, often only to fulfill what is known as "buy it, fix it, sell it" strategy (Christophers, 2022; Fields & Uffer, 2016). This mode of housing provision (the 'securitization mode'), was enabled by the development of securitization legislation and the opportunities this indirect ownership of real estate assets affords to go beyond the highly inefficient primary market through which first-time capital is usually raised (Hardin III & Wu, 2010). This scaling up of property investors in this next mode of housing provision has created a disjuncture between the meaning of property as home and the investment it has become, a dynamic clearly evident in the US (Ashton, 2018; Beswick et al., 2016; Fields, 2019). As noted above, access to traditional real estate investment products remains largely limited to institutional investors and high-net-worth individuals. This is often due to the high costs and complexities involved in securitization—constraints that can be further exacerbated by fragmented and bank-dominated housing finance systems, as is the case in the EU (see, for example, Roig Hernando, 2017). Investments therefore must be large enough to offset these expenses, creating a significant barrier for smaller investors. These constraints open a space for actors who can strategically harness tokenization, connecting the consumer side, where retail investors are often excluded from traditional real estate markets, with developers seeking more

efficient, lower-cost financing.

By bridging this divide, tokenization could thus be used to expand securitization, enabling another mode of housing provision, especially in niche markets like Luxembourg. In this way, tokenization could be understood as a horizontal expansion of the securitization mode. It is thus crucial to understand how tokenization functions and what purpose it serves. Researching tokenization is nonetheless complicated by a shifting regulatory environment—we thus now turn to the way in which we approached this topic empirically.

### 3. Methodology

This empirical investigation began as part of the Applications of Blockchain Technology in Land Administration Systems, funded by The Luxembourg National Research Fund (FNR) in 2021 (Phase I; see Table 1), which examined the potential impact of state-led blockchain adoption in housing conveyancing and mortgaging across several European jurisdictions (Proskurovska, 2023b, 2023a; Proskurovska & Dörny, 2022). Luxembourg was one of the cases researched. This case showed a lot of promise: a consortium of state agencies had just launched one of the first public sector blockchains in the EU, aiming to boost public-private collaboration and blockchain innovation.<sup>4</sup>

Phase I focused on mapping housing-transaction processes, to understand how existing housing-transaction routines articulate the supply and demand sides of housing provision, how these routines are organized, how they might be reconfigured through blockchain adoption, and which actors could be affected. To this end, the first author conducted desktop research<sup>5</sup> complemented by six semi-structured interviews with representatives from the Ministry of Digitalization, the Tax Authority, two housing researchers, and two non-profit blockchain promoting organizations. Analysis of these interviews and secondary materials were used to map organizational processes in housing transactions, represented as activity UML diagrams. By discussing these processes with interviewees, the research identified the market's key actors, their roles in conveyance organization, and how they might be affected by automation or blockchain integration.

These six interviews were complemented by roughly ten informal interviews with bankers, land-registry officials, real-estate developers, and other professionals encountered during the National Housing Week ('Home Expo', 9–17 October 2021) organized by the Ministry of Housing. The early interviews suggested that, while state-led blockchain initiatives had potential to streamline conveyancing, they ultimately focused on credential verification (diplomas, building permits). Interviewees also pointed to private-sector fintech innovations developing in Luxembourg, prompting a shift in research focus from the somewhat symbolic, state-driven initiatives to the private sector ventures enabled by recent updates to Luxembourg's securitization laws.

Notably, until 2021, Luxembourg's securitization industry depended on legacy information systems, with security registers playing a key role. In this context, custodians, that is institutions responsible for maintaining records of customers' securities, are vital. Custody of securitized assets involves not only protection but also oversight of virtual assets, including investor identification and compliance (ALFI, 2020, p.9). Ensuring this accountability incurs costs that require economies of scale to offset. Consequently, until recently, traditional securitization of small property stakes in Luxembourg by retail investors was economically

unviable. However, the legislative reforms of 2019 and 2021, known as Blockchain Laws I and II, established blockchain as a legally valid infrastructure for issuing and transferring securities, with blockchain-based tokens recognized as equivalent to book-entry or registered securities.

During the ABBL webinar "The Blockchain Laws in Luxembourg: Is the Market Ready?" on September 23, 2021, CMS Luxembourg partners highlighted the significance of these changes. Luxembourg's securities framework of 2001, recognizing bearer securities (physical certificates transferable by delivery) and registered securities (ownership recorded in issuer-maintained registers). In 2013, dematerialized securities replaced physical certificates with book entries held by service providers. The Blockchain Laws later built on this, complementing earlier initiatives like the CSSF's 2014 guidance on virtual currencies and authorization of crypto exchanges. Crucially, these laws allowed securities to be natively tokenized on the blockchain from inception, removing the need for traditional forms. The second law allowed any credit institution or investment firm authorized in a Member State of the EEA to act as central account keeper for unlisted debt securities, simplifying asset management.

These updates created new opportunities in Luxembourg's burgeoning but niche residential real estate market, paving the way for a reworking of the scale at which finance can penetrate real estate located in Luxembourg and beyond. Against this backdrop, the research followed one of the first private real estate investment initiatives domiciled in Luxembourg, hereafter the startup, which pioneered the use of blockchain to sell and manage programmable securities backed by residential real estate. Marketed as "an alternative pathway" to both property funding and property ownership, the startup became the focal point of the subsequent research phase (Phase II, 2022–2025).

Phase II continued the investigation as an independent, longitudinal qualitative study. Participants were recruited through a combination of purposive and snowball sampling, drawing on contacts from Phase I and from professional and industry events related to blockchain and tokenization. This phase aimed to understand how tokenization functioned in practice and to assess its potential impact on both sides of housing provision in a rapidly shifting policy landscape both in Luxembourg and at the European scale.

To do so, the authors built on their complementary expertise including economic geography, housing studies and urban geography. At this stage, we adopted an exploratory approach to the case, attempting to source what Gordon Clark calls stylized facts - "simple representations of the observed world drawn from empirical evidence and/or theory" (Clark & Wójcik, 2024). Many of these were obtained through empirical observation complemented by formal and informal interviews, with the latter conducted at tokenization-focused events in Luxembourg (e.g., Fintech Recruits, Belval; Real Estate Tokenization at Banque Internationale à Luxembourg). The Phase II sample comprised individuals familiar with the operations of the startup, who provided contextual insights into the organization and challenges of property tokenization. This phase included six additional semi-structured interviews (conducted between 2022 and 2025) with executives from the fintech, real estate, and information and communication technology sectors, as well as with investors: actors directly involved in structuring, managing, or supporting tokenization initiatives.<sup>6</sup>

This embeddedness in the research context aimed to deepen the understanding of housing tokenization processes and mechanisms on the ground. Interpretations developed in this context were triangulated

<sup>4</sup> [https://ctie.gouvernement.lu/fr/actualites.gouvernement2024+fr+actualites+toutes\\_actualites+communiqués+2019+05-mai+23-hansen-blockchain.html](https://ctie.gouvernement.lu/fr/actualites.gouvernement2024+fr+actualites+toutes_actualites+communiqués+2019+05-mai+23-hansen-blockchain.html)

<sup>5</sup> These include government and EU policy documents, Big Four consultancy reports, white papers from tokenization platform provider and other blockchain infrastructure providers; briefs from major local legal firms, dedicated webinars, and media coverage of blockchain and housing innovation in Luxembourg.

<sup>6</sup> Due to the limited size of Luxembourg's professional and policy community, and the close interconnections among actors in the blockchain and housing sectors, providing further identifying details or assigning unique identifiers to interviewees could inadvertently compromise participant anonymity. To mitigate this risk, only sectoral descriptors are reported, and interview data are cited solely by year, without individual numbering.

**Table 1**

Data collection overview.

| Phase | Period             | Focus                                                                 | Interview count, Type            | Medium & Duration & Place                           | Themes                                                                                                                                                                                                                                          |
|-------|--------------------|-----------------------------------------------------------------------|----------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I     | Aug - Oct 2021     | Housing transactions as is and to be & state blockchain initiatives   | 6 semi-structured and 8 informal | In-person / online, 30–120 min; Luxembourg          | The organization of Luxembourg's housing and real-estate market; existing conveyancing routines and their bottlenecks; perceived potentials and risks of blockchain-based upgrades to land administration systems; key market players, etc.     |
| II    | Oct 2022 -Jan 2025 | Regulatory changes/<br>Tokenization of real estate;<br>Company#1 case | 6 semi-structured and 4 informal | Online / in-person, 30–60 min<br>Canada, Luxembourg | Legal/technical setup and structuring of tokenized real-estate vehicles; costs; token design and operation; behavior of market participants; perceived risks, opportunities, and regulatory challenges associated with real-estate tokenization |

by drawing on corporate financial records, records of property transactions and a desktop analysis of secondary literature on evolving blockchain regulations and on the specificities of the Luxembourg housing market. Given the difficulty of stabilizing an interpretation, we moved away from an assessment of the prospective impact of the venture itself to a broader reflection on the change tokenization brings to the finance-housing nexus.

By adopting a longitudinal, qualitative exploratory research design, we were able to follow part of an ongoing process that has become a thorny tokenization path: from the opportunistic use of new regulation geared towards the financial sector, to the development of a new ecosystem and, at present, what appeared to be a regulatory deadlock. What we offer here is a description of an attempt at housing tokenization in a country that, because of its small size and economic specialization, has a more permeable boundary between the financial sector and other economic sectors. This makes more visible the ways in which local real estate professionals have attempted to extend securitization of a fund-like structure through tokenization to solve a local housing provision problem. We now move to a description and critical assessment of the startup's attempt to harness the tokenization of housing.

#### 4. Tokenizing 'homeownership': can it enable a new mode of housing provision in Luxembourg?

Luxembourg, though one of the smallest countries in the world, is the second-largest fund domicile globally, after the United States, with over €5 trillion in net assets under management (CSSF, 2025) and offers advanced business services, provided by a dense network of international law firms, auditors, and consultants, as well as regulatory structures tailored to various fund types and international investors. Luxembourg secured its competitive edge by swiftly embracing and capitalizing on the opportunities presented by increasingly international financial markets (Dörny, 2016). This proactive strategy helped establish the country as the preferred domicile for both UCITS (Undertakings for Collective Investment in Transferable Securities), targeting public, retail investors seeking diversified and regulated investment products across EU borders, and Alternative Investment Funds (AIFs), which serve private investors through vehicles such as private equity, real estate, and hedge funds (Terblanche, 2016). Following this tradition, Luxembourgish lawmakers were among the first in the EU to adopt blockchain-friendly legislation, starting with Blockchain I (2019) and II (2021), which recognized the use of distributed ledger technology (DLT) for the issuance and transfer of dematerialized securities, followed by Blockchain III (2023) to implement the EU DLT Pilot Regime, and most recently Blockchain IV (2024), which formally recognized DLT as a valid infrastructure for issuing both debt and equity instruments.

Yet this prosperity comes with a steep cost of living and deepening affordability issues, with housing prices being a major concern. With the average cost of a house nearing €1 million (Observatoire de l'Habitat, 2025), homeownership is increasingly out of reach for much of the population. For example, a significant portion of Luxembourg's workforce lives beyond its borders; 47% of those employed in the country are

cross-border workers commuting from neighboring France, Belgium, and Germany (Statec, 2025). At the same time, the vision of universal homeownership, a cornerstone of the country's housing policy for most of its modern history, has taken a blow. According to Eurostat, this rate has fallen by nine percentage points in the last decade (from 72.5% in 2014–63.5% in 2024). Researchers explain that as Luxembourg's financial industry expanded, and with a population largely composed of migrants, the complexity of accessing residential land also increased (Observatoire de l'Habitat, 2021; Paccoud et al., 2021). As a result, housing supply has failed to keep pace with a growing population and house prices more than doubled between 2010 and 2022 (Eurostat, 2025). This marked a shift to the buy-to-let mode of housing provision when, in an isolated effort to boost housing supply in the early 2000s, several tax incentives were introduced to create a new outlet for off-plan properties, and thus in turn stimulate the supply of housing (Mezaroš & Paccoud, 2022). These measures had little effect on overall housing supply, with the number of completed dwellings stable since the 1970s (STATEC). Instead, developers seem to have redirected the properties built for owner-occupiers towards buy-to-let investors, who were able to purchase properties faster (no chain transactions, at the pre-financing stage) and in greater quantity than would be owner-occupiers (Mezaroš et al., 2025).

However, the hike in interest rates decided by the European Central Bank in 2022 reduced potential buyers' purchasing power by 20–30%, disrupting the buy-to-let mode of housing provision in Europe. In niche real estate markets like Luxembourg, where mortgage lending is primarily funded through traditional bank deposits, the number of new mortgages granted decreased (D'Alessandro, 2024). In 2023, home mortgages totaled just under EUR 4 billion, a 44% drop from EUR 7.18 billion in 2022, marking the lowest level in over a decade. Compared to previous years, the number of transactions declined by 52% from 2021, while the total financial volume fell by 53% (ibid.). As a result, large local developers have preferred to withhold new properties from the market, aware they can no longer sell them at the usual prices citing evaporated housing demand to explain such a strategy. The number of transactions of newly built apartments has decreased by 72.8% between the last quarter of 2022 and the last quarter of 2023, while the adjustment in the price of off-plan apartments is half that of existing apartments (Observatoire de l'Habitat, 2024). The exhaustion of the buy-to-let mode of housing provision pushed housing operators to look for new sources of both finance and housing demand. Close ties between real estate and financial actors, combined with the early adoption of favorable legislation, have drawn the attention of local real estate professionals to the potential of housing tokenization.

##### 4.1. Exploring tokenization mechanisms beyond the elimination of transaction costs and intermediaries

In 2021, the founders of the startup incorporated 'SV S.A.', an unregulated securitization vehicle (SV) structured as a public limited liability company (hereinafter SVSA), to raise around 8 million euros. The capital was intended to finance the acquisition of shares in another

company structured as an S.A.R.L. (hereinafter Company#2), which at the time directly owned a high-end, multi-unit residential property under construction, Property #1<sup>7</sup> (see Fig. 1). According to the SVSA offering memorandum issued in 2021, should the fundraising be successful and once Property #1 was completed and either rented out or sold, the issuer would receive the revenues generated by the asset, while Class B shareholders would be entitled to dividends from any profits realized.

At first glance, the startup is a fund-like structure resembling that of traditional private, non-listed investment vehicles such as non-traded REITs. These entities pool capital through securities offerings to acquire income-generating properties and are typically not listed on public stock exchanges. People investing in these vehicles receive returns derived from rental income and potential capital appreciation (Henderson et al., 2016). These are paid out as dividends after deducting fees covering operational and management costs. Like the REITs in the United States, SV vehicles in Luxembourg benefit from a largely tax-neutral regime: although subject to the standard corporate income tax rate (around 24.94% when including the Luxembourg City municipal tax), a securitization company can deduct all obligations, such as dividend commitments to shareholders or interest payments to bondholders, from its taxable income. This means only the remaining net profit, if any, is taxed, allowing for a reduction in the overall tax burden compared to direct property ownership.

But the startup differs from a fund in a number of ways. First of all, within Luxembourg's legal framework, SVs like SVSA do not possess independent legal personality in the traditional sense; they do not directly operate businesses or manage assets. Instead, they rely on designated service providers, or operation companies, to act on their behalf. These service providers are typically responsible for issuing financial instruments and may also initiate segregated compartments within the SV. Each compartment can be used to isolate specific assets and liabilities, operating independently, even in cases of insolvency, thus "ring-fencing" risk for investors (Loyens & Loeff, 2022). This setup allows SVs to issue a range of instruments, from equity-like shares to debt securities, across compartments tailored to different investment profiles.

Holding 100% of SVSA's A-class (controlling) shares, the startup's Operating Company (thereafter OpCo), a public limited company incorporated just days before SVSA to develop and promote a trading platform for real estate and related activities, held significant influence over the vehicle's structure and strategic direction, including the potential creation of new compartments. The OpCo's founding parties also sat on SVSA's board of directors, and a contractual agreement between the two entities tasked the OpCo with overseeing both management and tokenization, entitling it to service fees equal to 1.5% of the nominal value of subscribed securities. As one interviewee explained, the long-term vision was for investors to "own one part of this building, but also of the next building," ultimately holding shares "representing all the buildings" in a diversified portfolio [Interview, 2021]. This model suggested a dynamic issuance strategy that could use compartmentalization to both spread investor risk across multiple real estate assets and tailor offerings to different investor profiles such as real estate developers.

Indeed, the startup positioned its undertaking as a service to both professional developers struggling with declining demand and rising financing costs, and new and existing retail investors deterred by soaring housing prices, thereby proposing a new mode of housing provision through the tokenization of residential real estate. Thus, unlike traditional non-public funds which cater primarily to institutional or

qualified investors, this venture targeted smaller retail investors, specifically natural and legal persons who do not qualify as "professional investors" under EU rules (SVSA Offering Memorandum). The minimum investment was set at just EUR 1000, a figure below the standard entry point for comparable investment structures. SVSA investors do not "have to make savings for five to ten years before [they] can buy [real estate] and instead can start climbing the property ladder by gradually accumulating equity immediately" [Interview, 2021].

The startup backers also claimed that investors were offered priority leasing rights, entitling them to rent units ahead of other prospective tenants navigating the overheated rental market. This approach supported their broader vision of "a community", where some participants would live in the property while others invested for rental income. As one interviewee explained, "the owners of the building will use the building," dissolving the traditional divide between landlords and tenants [Interview, 2021]. Instead, both groups would be integrated "so, there will be no dichotomy (...) 'you [won't have] one side [as] the owner, [and] on the other side, the users.'" [Interview, 2021]. Should a renter-owner-investor wish to live in the property 20–30 years down the line, they could, by gradually accumulating a sufficient ownership stake, effectively reside there at no net cost: their rental payments would flow back to them as dividends tied to their shareholding. "Living as a service is the big vision (...). [You can] have thousands of dwellings (...) [so] people can buy in, live in New York, then move to Singapore, [and so on]". They pay for what they use and get back what they invested." [Interview, 2024]. In other words, if sometime in the future an investor decides to "leave" [property#1], "the sale will be done from one member of the community to another" [Interview, 2024].

By pooling capital from retail investors, the model enabled more affordable financing for property developers, who, as in the case of aspiring homeowner-occupiers, often had little choice but to rely on traditional mortgage banking. A developer relying on this model might "say, okay, I have 100 apartments. Instead of following the classical way, I propose my development to [the] OpCo" and thus build without incurring "a lot of costs, [and] a lot of debt." [Interview, 2024].

Blockchain technology was the key enabler of this vision, setting the startup apart from traditional private real estate fund structures. Indeed, both the investors interested in purchasing the securities, and eventually the developers that might be interested in selling their properties to the community of *token holders* were required to do so via OpCo's platform underpinning the issuance, transfer, and management of the shares that SVSA issued in the form of tokenized securities.

The platform's core was a blockchain-based digital identity issued by Luxembourg fintech Company#3, which specializes in providing an alternative digital infrastructure for the issuance, transfer, and management of tradable digital assets/security tokens, including tokenized loans and structured notes. This digital identity is designed for secure and compliant tracking of token ownership in line with Luxembourg's securitization and AML regulations. In practice this means that once an investor is whitelisted, their digital identity links to a digital wallet created during subscription, enabling them to interact with their tokenized security. Because these digital identities are implemented as smart contracts, the OpCo, as platform owner, can ensure that customized permissions and compliance rules embedded in the security tokens automatically prevent unauthorized or non-compliant transactions directly on-chain.

This setup streamlined onboarding while keeping it closely aligned with the formal subscription process for registered investors in a public offering. Ownership of the SVSA securities represented by tokens remained recorded in the issuer's share register, which was updated in sync with transfers on the startup's platform. By automating compliance controls, the OpCo was able to streamline both issuance and post-issuance processes, which Company#3 claims has reduced compliance and administrative costs by nearly 90 %. Additionally, the platform allowed the startup's investors to exercise voting rights attached to their securities, conduct secure peer-to-peer token transfers, and make

<sup>7</sup> OpCo's Class A shares are equally owned by Company #4 S.A.R.L. and Company #5 S.A.R.L. Company #4 S.A.R.L. is owned by Company #6 S.A.R.L., which also holds 25% of Company #2, alongside Company #7 (25%) and Company #5 S.A.R.L. (50%) – as we learned by consulting Luxembourg Business Registers in 2025.

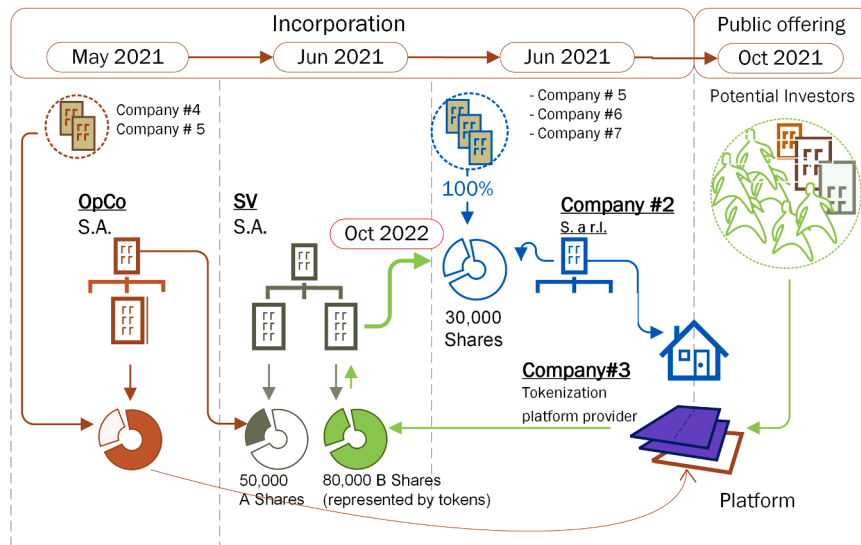


Fig. 1. The group until the completion of the share deal. Source: Illustration by the first author.

payments directly between wallets. With due diligence procedures encoded in smart contracts and executed only when predefined conditions were met, these transactions could bypass traditional intermediaries involved in both direct and indirect homeownership transfers, potentially bringing the vision of the integrated community of token holders even closer: *“For me the banks are at the end of their life (...). The next generation will live in a community system. It will be part of the shared economy. Instead of going through a bank - which creates a lot of obstacles - I’ll go to my community and say, ‘I need 10,000 euros to buy a car.’ And people will say, ‘Okay, yes, we understand. We’re part of the same community. You’ve already borrowed money and paid it back before. We’ll give you the 10,000 euros, and we’ll take two percent.”* [Interview, 2021].

If tokenization is seen as a new mode of housing provision, its essential characteristic is a horizontal expansion of indirect property ownership - mirroring the spread of homeownership in the latter half of the twentieth century. Whether this new horizontal expansion offers a way out of the inequities of previous provision modes remains an open question. What is more certain is that actual tokenization seems further away than both instigators and commentators originally imagined.

#### 4.2. Revealing the limits of the extension of securitization via blockchain: can we really tokenize homes?

While horizontal expansion may well become a reality in the future, the case of this Luxembourgish venture offers little indication of a radical change in housing provision. Indeed, the offering campaign has managed to attract around 600 subscribers in the two months following the launch of its offering, which eventually resulted in approximately 350 active token holders. Table 2 below shows the distribution of

investors by age group and the amount invested.

Additionally, the startup also indicated that 79% of these investors have their residence in Luxembourg. These broad characteristics of the investors thus mirror those found for ‘traditional’ buy-to-let investors in Luxembourg: they are mainly over the age of 45 and born in Luxembourg (Mezaroş & Paccoud, 2022).

On the supply side, while Property #1 was purchased and rented out, traditional banking was hardly removed from the equation. In fact, the subscription note that complemented SVSA’s Offering Memorandum built the involvement of debt financing into the model by design. Indeed, while the Offering Memorandum merely reserved the right to cancel the issuance if financing was not secured, the Subscription Note made it explicit that the issuance of securities would only proceed if (i) at least EUR 4 million was raised through the offering and (ii) the loan was granted to the issuer by July 1, 2022.

This introduction of a hybrid financing approach can be explained by the startup’s ambition to acquire more properties quickly and diversify its investment portfolio, an ambition they were actively pursuing at the time of the Offering Memorandum. In 2024, OpCo announced that SVSA’s portfolio was growing: following a joint decision of “the community,” SVSA was set to acquire a second residential building in one of the central districts of Luxembourg City. This deal, however, ultimately fell through. As one interviewee explained, *“[the project] had to stop because of the lack of finance... the banks put a hold on the credit lines [the startup] had, and they needed more money from the [token holders].”* [Interview, 2024]. Although onboarding for the property had already begun and two funding rounds were launched yielding some success, many investors hesitated. *“People were worried about the prices... they said real estate is not a good investment and wanted to wait.”* Despite a “decision

**Table 2**  
The general distribution of investor characteristics (as of 8 July 2024).

| Age Group    | Number of Investors | Total Amount Invested (Euro) | Average Amount Invested (Euro) | Share of Invested Total | Number of Subscribers | Total Monthly Subscription Amount (Euro) |
|--------------|---------------------|------------------------------|--------------------------------|-------------------------|-----------------------|------------------------------------------|
| 19–25 years  | 18                  | 83,413                       | 4634                           | 3%                      | 3                     | 477                                      |
| 26–35 years  | 119                 | 644,412                      | 5415                           | 23%                     | 29                    | 8847                                     |
| 36–55 years  | 173                 | 1249,224                     | 7221                           | 44%                     | 41                    | 10,267                                   |
| 56 + years   | 34                  | 833,596                      | 24,518                         | 30%                     | 15                    | 5565                                     |
| <b>Total</b> | <b>344</b>          | <b>2,810,645</b>             | <b>8,170</b>                   | <b>100%</b>             | <b>88</b>             | <b>25,156</b>                            |

Source: Company #1, personal communication; permission on file. Information on three token holders aged 18 or under were not included in the totals. The total amount invested includes the sums contributed via the monthly subscription up to the cut-off date.

from the general assembly to buy it," the funding gap proved insurmountable. Attempts to renegotiate with the property developer were unsuccessful, and "in the end, the board decided not to buy it." [Interview, 2024].

These developments highlight not only the limits of housing tokenization in achieving true decentralization and democratization, but also the constraints of blockchain in extending securitization. Traditional financial institutions continue to play a critical role in executing the model. Rather than displacing banks, the platform integrates them into a broader framework of shared ownership and tokenized participation.

Similarly, it does little to blur existing dichotomies, as there appears to be limited interest, either from [token holders] or tenants, in bridging the traditional divide between owners, renters, and investors. There was little interest among the token holders to become tenants in Property#1, as "most were living in Luxembourg and weren't interested in committing to something" [Interview, 2024]. Interestingly, Property#1 tenants have also shown little interest in becoming part of token holding community. When offered the option to cover rental guarantees - typically equivalent to two months' rent in Luxembourg - by purchasing tokens instead of depositing cash, they declined: it remains unclear "why they didn't do it (...) tokens - that's still your money, it is your value." [Interview, 2024].

However, it seems that the greatest obstacle to transforming housing provision remains the illiquidity of tokenized homes. As of January 2025, there was no meaningful secondary market for SVSA securities. Efforts to address this challenge have been further complicated by the EU-wide MiCAR. Notably, MiCAR explicitly excludes security tokens from its scope, classifying them instead as transferable securities or other financial instruments under MiFID II. As a result, they fall under the full scope of EU securities law, including the Prospectus Regulation, licensing requirements, and post-trade obligations set out in the Central Securities Depositories Regulation (CSDR). The second phase of MiCAR, which took effect on December 30, 2024, also introduced stricter licensing rules for Crypto Asset Service Providers (CASPs), adding further regulatory hurdles.

Crucially, these legal constraints did not begin with MiCAR (we will return to this issue shortly). Even prior to its adoption, OpCo was unable to legally facilitate token exchanges on its platform. As one interviewee explained, "That's not possible because, for that, you need an agreement [a license from the regulator] which is very, very complicated to get." [Interview, 2024]. For this reason, the venture planned to rely on an infrastructure provider that was, at the time, developing a so-called "brokerage platform," which one token holder described as a "billboard." The platform would simply display interest from potential buyers and sellers without actively brokering transactions. Essentially, it signals a buyer's interest so that sellers can initiate discussions about price if they choose to engage. Under these settings, the platform would only show the current token price based on professional valuations conducted every six months (e.g. one euro per share), while all other terms would be left open to negotiation between the peer shareholders: once they reached an agreement, the outcome would be communicated to the OpCo, who would then record it on blockchain. However, to avoid being classified as a broker, the process could neither be automated nor involve transaction fees. This raised fundamental concerns about the startup's original business model, which relied on revenue from onboarding and exchange fees. The team later learned from the regulator that if the OpCo charged fees, the platform would be considered a stock exchange, requiring additional licenses and prohibiting any involvement in transactions. Ultimately, this left advertising as the only viable model to enable a secondary market for the community of token holders. In the meantime, when members wish to sell, other community members typically buy the tokens, or they are covered through monthly subscriptions, an alternative mechanism that also offers the option to buy into the SVSA stake established by the company.

However, as mentioned above, the MiCAR has further complicated the promise of democratizing real estate investments through

tokenization: from now on tokenized securities are essentially digital representations of traditional financial instruments and trading them requires compliance with existing regulatory frameworks, such as obtaining licenses to operate as an investment firm or multilateral trading facility. As one interviewee explained, these requirements make targeting retail investors much more difficult: "It's not because you tokenize that you can suddenly reach anyone you want. There are securities laws. In my opinion, they are stupid. If you are not rich enough, you cannot invest" [Interview, 2025]. More importantly, as it also regulates the so-called "cash leg" that can be potentially involved in the exchange of tokenized securities, i.e. stablecoins, clients of service providers like Company#3 would luckily require additional licensing to remain compliant.

This effectively makes the promise of the 24/7 tradability of tokenized securities, let alone tokenized housing, out of reach for smaller players. New European-wide legislation that imposes stricter controls on secondary trading and investor eligibility casts doubt on the viability of business models described above and the broader promises of extended securitization and democratized investment. Our investigation of the ongoing development of tokenization in Luxembourg thus shows that it cannot yet be described as a new mode of housing provision.

## 5. Discussion and outlook

This ongoing empirical engagement with the unfolding story of the Luxembourgish startup offers valuable insights into how housing tokenization operates in practice: its mechanisms, its ambitions, and, perhaps most importantly, its limits. We now build on these insights to draw more general conclusions about the extent to which blockchain can reshape the way finance penetrates residential real estate.

At first glance, the startup appears to exemplify the potential of tokenization to overcome some of the core limitations of conventional real estate securitization, and in doing so, to challenge prevailing political assumptions about who is entitled to make social claims on housing assets and the future revenues they generate. More specifically, by fragmenting risk into small, manageable bundles and leveraging an unregulated special purpose vehicle, the startup has lowered entry barriers for real estate investors, mostly local residents, thus allowing even the smallest investors to participate in private investment. These investors, shielded from liabilities beyond their stake, could earn income from a property, much like shareholders in a closed-end REIT, but without the burdens of high capital requirements.

Indeed, its B-share holders were granted a say in asset-level decisions, embedded in their tokenized securities through SC, offering a level of influence typically reserved for large institutional investors or high-net-worth individuals accessing such assets via private finance. That such governance rights were extended to ordinary retail participants, without the long-term obligations or embedded relationships of mortgage-based owner-occupation, suggested a reimagining, albeit largely symbolic, given the small scale of the venture, of how decision-making power might be distributed in tokenized residential real estate.

However, such involvement offers limited protection for small investors against corporate actors seeking capital to fund large property development projects, and to avoid the pitfalls faced by crowdfunding ventures discussed in [Section 2.2](#). It is important to note that the OpCo's platform business model explicitly targeted developers, focusing on the supply side. By directly involving developers and connecting them with an ostensibly boundless crowd of retail investors, the startup's founders aimed to create more than a "self-sustaining ecosystem." The goal was to free residential property developers from the dependence on local banks or the uncertainties of presales and to thereby enable more flexible financing. Nevertheless, as with early crowdfunding efforts, the risk ultimately rested with the token holders rather than the developers.

Notably, two of the three beneficiaries holding Class A shares in the OpCo also controlled Company#2, the developer of Property#1, which the OpCo, the platform's de facto owner, later acquired on behalf of the SVSA.<sup>8</sup> In other words, the startup was designed to operate as a creative risk-transfer mechanism that limited risks for developers and OpCo founders while opening the door to an essentially unlimited supply of cheap finance. This, however, depended on the OpCo's ability to successfully expand its range of investible assets, grow its community of investors by integrating geographically dispersed participants into a single platform, maintain investor confidence, and monetize participation through onboarding and exchange fees.

Now, drawing broader conclusions, we would argue that this case illustrates that housing tokenization has the potential to open up new modes of housing provision and indeed to democratize real estate investment. However, this promise does not arise from the potential of blockchain enabled tokenization to transform the fundamental processes through which global capital (crypto or otherwise) and the materiality of the niche Luxembourg housing market are integrated. In practice, it neither meaningfully extended securitization nor displaced traditional mortgage lending facing familiar structural constraints: from the outset, the startup relied on bridge loans covering half the capital, even as its founders departed from their typical roles as real estate developers to act instead as community builders and platform operators.

More importantly, despite its ambitious vision, one key feature of traditional securitization is still missing: a functional secondary market, unless a third party such as a bank, is involved. Without it, the core value proposition of securitization, liquidity, has yet to materialize. Token holders cannot freely trade their claims *peer-to-peer*; instead, they are effectively locked into the platform's ecosystem, a contained marketplace whose reliance on blockchain contributes to its disconnect from broader capital markets. Consequently, investing in assets configured to circulate on these platforms entails lock-in effects similar to those seen in earlier fundraising models described as platform capitalism (Langley & Leyshon, 2021). At the same time, these ventures replicate the existing system in which housing provision is primarily funded through bank-held deposits from individuals and businesses.

One could certainly argue that this outcome has been shaped by external factors beyond the startup's control, such as interest rate hikes, a cooling housing market, and regulatory uncertainties, which eventually become hurdles. However, for those experimenting with alternative housing finance models, this result is not merely underwhelming: it is structurally limiting. As it stands, unless SVSA's investors' tokens are repurchased by others within the community, they hold assets with the commitment and immobility of homeownership, ironically, the very condition tokenization was meant to transcend. Against this backdrop, the formalization of tokenization as part of traditional finance—reflected, for example, in the recent BIS report which defines tokenization as “*the generation and recording of a digital representation of traditional assets on a programmable platform*” (BIS, 2024, p. 4), and in the EU's MiCAR, which, despite introducing certain obstacles, also provides greater regulatory clarity—looks like a good news for financial incumbents. By framing tokenization not as a transformative force, but as a technical add-on: an auxiliary layer that enhances legacy financial infrastructure rather than disrupts it, this formalization signals yet another round of reintermediation (Proskurovska & Birch, 2025).

As we write this, housing tokenization endeavors aimed at extending securitization and thereby creating an alternative source of funding, are still under construction. The goal is to make tokenized homeownership more comparable to other, more conventional financial assets, but its outcomes and broader impact currently remain uncertain. Given recent regulatory developments in the EU, we argue that, for now, there is little evidence that housing tokenization is a next step or distinct path for advancing the financialization of housing in this part of the world.

Simply put, tokenization does not fundamentally alter the way finance penetrates real estate markets. On the contrary, traditional lenders remain central players, raising questions about democratization potential of tokenization. As our case demonstrates, the number of investors involved is negligible, mostly local, and includes only a small minority of younger individuals, a demographic largely excluded from both the housing market and private market investments.

Still, as the roles of financial incumbents in these emerging practices are being reconfirmed, it becomes ever more crucial to shed light on how tokenization unfolds in practice, who shapes this process, and for what ends, as well as how it evolves. Crucially, as this article shows, tokenization has the potential to extend securitization and, more importantly, to blur the traditional divide between owners, renters, and investors, as well as between direct and indirect forms of property ownership, with potentially far-reaching consequences for the future of housing and housing finance. For example, Proskurovska and Morawski (2026) show that, when tokenization is operationalized through blockchain-based conveyancing systems, it can dramatically increase housing market liquidity, amplify price volatility, and ultimately alter the characteristics of real estate as an asset class. This implies that tokenization may eventually bestow directly owned properties with characteristics that erode the stabilizing role of illiquidity in real estate, thereby rendering housing price dynamics more akin to those of stock exchanges and opening up new frontiers of financialization.

One of the main challenges here is how to study this new wave of reintermediation. In this research, we adopted a longitudinal approach to track developments as they unfolded in real time and to extract stylized insights grounded in corporate practice. Without direct empirical engagement and dialogue with the corporate actors and investors involved, scholars risk reducing their analysis to simplistic dichotomies of failure or success, benefits or shortcomings, while overlooking the material effects that unfold in real-world contexts. As tokenization continues to evolve, this engagement can help us to better understand both its mechanisms and long-term social impact, especially for marginalized groups.

This case also underscores the urgency of expanding both the involvement and the range of actors working to shape the trajectory of tokenization. In particular, we call on policymakers and other housing stakeholders to engage more directly with tokenization and to match the level of attention already afforded to it by actors in the financial and real estate sectors who are actively shaping its mechanisms, asset configurations, and markets. Such engagement is crucial, as tokenization is increasingly leveraged to reconfigure familiar arrangements in housing provision, often framed through narratives of democratization, participation, and community empowerment that also serve to rebalance the distribution of risks. Early and more hands-on participation from these actors could play a decisive role in steering the future development of tokenization in real estate and, in turn, its effects on a strategic sector such as housing, especially as tokenization comes of age, reshaping the mechanisms, scale and forms through which finance penetrates housing provision.

#### CRediT authorship contribution statement

**Anetta Proskurovska:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Antoine Paccoud:** Writing – review & editing, Writing – original draft, Validation, Investigation.

#### Declaration of Competing Interest

The authors, Anetta Proskurovska and Antoine Paccoud certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants;

<sup>8</sup> see footnote 1 in Section 4.1. page on X)

participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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